

HoldMeTolt — Master Feature Specification

Document Version: 2.0

Source of Truth: Repository Root — Authoritative Product Feature Catalog

Rule: Every screen, user flow, and functional capability across all product phases is detailed here.

Last Updated: 2026-09-05

1. Phase Legend & Delivery Roadmap

All features are strictly tagged with their planned deployment phase:

Phase Tag	Phase Name	Target Window	Core Focus
[P0]	MVP Core	~10 Days	Spreadsheet replacement: self-logging in HH:MM:SS , live scoreboard, mandatory goals, auto-punishments, PFP download, 1-click Discord summary.
[P1]	Automation & Integrations	2 Weeks	YPT API Bot automated ingestion, 24/7 @Ho1dMeToItBot Discord bot, auto-expiring countdown timers.
[V1]	Gamification & Fair Math	2–3 Weeks	Overflow diminishing returns ($\alpha = 0.5$), Auto-balancer snake draft (15h cap), daily challenge streaks, achievement badges, member profiles.
[V2]	Spontaneous Duels & Scale	2 Weeks	Peer-to-peer 1v1 sprint duels (mutual consent, zero mod overhead), multi-guild schema readiness.

2. Master Feature Catalog (Quick Matrix)

Feature ID	Feature Name	Module	Phase	Target Persona
FEAT-AUTH-01	Discord OAuth 2.0 Authentication	Auth & Identity	[P0]	Participant, Admin
FEAT-AUTH-02	Public Read-Only Spectator Mode	Auth & Identity	[P0]	Spectator / Public
FEAT-AUTH-03	Multi-Guild Tenant Isolation	Auth & Identity	[V2]	System
FEAT-CHAL-01	Multi-Format Challenge Creator (Team, Duo, Solo)	Challenge Ops	[P0]	Admin
FEAT-CHAL-02	Host Manual Event Kickoff Trigger	Challenge Ops	[P0]	Admin
FEAT-CHAL-03	Automated Countdown Timer & Expiration	Challenge Ops	[P1]	System
FEAT-CHAL-04	Auto-Balancer Snake Draft & 15h Cap	Challenge Ops	[V1]	Admin
FEAT-CHAL-05	Event Lock & Freeze Final Results	Challenge Ops	[P0]	Admin
FEAT-DECL-01	Declared Target Hours (HH:MM:SS)	Declarations	[P0]	Participant
FEAT-DECL-02	Mandatory Weekly Goals Checklist	Declarations	[P0]	Participant
FEAT-DECL-03	Pre-Kickoff Declaration Lock	Declarations	[P0]	System
FEAT-DECL-04	Host Goal Unlock & Mid-Event Edit	Declarations	[P0]	Admin
FEAT-LOG-01	Daily Clock-Time Self-Logging (HH:MM:SS)	Study Logging	[P0]	Participant
FEAT-LOG-02	24-Hour Single-Day Limit Validation	Study Logging	[P0]	System

Feature ID	Feature Name	Module	Phase	Target Persona
FEAT-LOG-03	Direct YPT API Bot Ingestion Endpoint	Study Logging	[P1]	System / Teammate Bot
FEAT-LOG-04	Admin Inline Hours Override Grid	Study Logging	[P0]	Admin
FEAT-LEAD-01	Head-to-Head Live Match Scoreboard	Standings & Math	[P0]	All Users
FEAT-LEAD-02	Unified Roster Standings Table	Standings & Math	[P0]	All Users
FEAT-LEAD-03	Dynamic Daily Catch-Up Deficit Engine	Standings & Math	[P0]	Participant
FEAT-LEAD-04	Overflow Diminishing Returns Scoring Engine	Standings & Math	[V1]	System
FEAT-PUN-01	Dual-Failure Auto-Flagging Engine	Accountability	[P0]	System
FEAT-PUN-02	Punishment Wall & Deficit Roster	Accountability	[P0]	All Users
FEAT-PUN-03	Direct Punishment PFP Asset Download	Accountability	[P0]	Flagged Member
FEAT-PUN-04	Host Pardon / Excuse Override	Accountability	[P0]	Admin
FEAT-PUN-05	Hall of Accountability Historical Archive	Accountability	[V1]	All Users
FEAT-DISC-01	1-Click Formatted Markdown Summary Copy	Discord Broadcaster	[P0]	Admin
FEAT-DISC-02	24/7 @Ho1dMeToItBot Slash Commands	Discord Broadcaster	[P1]	Discord Users
FEAT-DISC-03	Automated Daily & Final Results Embeds	Discord Broadcaster	[P1]	Discord Channel

Feature ID	Feature Name	Module	Phase	Target Persona
FEAT-GAME-01	Daily Challenge Streaks	Gamification	[V1]	Participant
FEAT-GAME-02	Condition-Based Achievement Badges (5 Types)	Gamification	[V1]	Participant
FEAT-GAME-03	Member Profile Cockpit (/profile/[id])	Gamification	[V1]	All Users
FEAT-DUEL-01	Spontaneous 1v1 Mutual Study Sprint Duels	P2P Battles	[V2]	Discord Members

3. Module Specifications: Authentication & Identity

§3.1 Discord OAuth 2.0 Authentication [P0]

- **User Flow:** User visits landing page → clicks "Login with Discord" → grants `identify` scope → redirected back with session.
- **Data Captured:** Discord Snowflake ID, username, global display name, avatar URL hash.
- **Persistence:** Upserts record into `User` table; synchronizes avatar and display name on every login.
- **Access Roles:** Default role `PARTICIPANT` . First user or server admin flagged as `ADMIN` .

§3.2 Public Read-Only Spectator Mode [P0]

- **User Flow:** Anyone visiting `/challenge/:id` without an active session can view the live match scoreboard, participant hours, and download the punishment PFP.
- **Security Invariant:** Write actions (logging hours, checking goals, editing rosters) are disabled and hidden for unauthenticated guests.

§3.3 Multi-Guild Tenant Isolation [V2]

- **Description:** Adds optional `guild_id` to `Challenge` and `User` models, preparing the codebase to be installed across multiple independent Discord communities without data leakage.

4. Module Specifications: Challenge Operations & Lifecycle

§4.1 Multi-Format Challenge Creator [P0]

- **Route:** /admin/challenges/new
- **Supported Formats:**
 - i. TEAM_VS_TEAM : Two named rosters (e.g. *Bees vs Butterflies*).
 - ii. DUOS : Pairs of $N = 2$ accountability partners.
 - iii. SOLOS : Free-for-all individual leaderboard ($N = 1$).
- **Configuration Fields:** Title, Start Date/Time, End Date/Time, Format, Team Names/Colors, Participant Assignment, Punishment PFP image URL/asset.

§4.2 Host Manual Event Kickoff Trigger [P0]

- **Mechanism:** Even after the scheduled start time arrives, the host retains a "Start Event Now" button to verify all rosters and declared goals before locking inputs.
- **State Transition:** Moves challenge from `UPCOMING` to `ACTIVE` . Locks all target hours and goal descriptions.

§4.3 Automated Countdown Timer & Expiration [P1]

- **Mechanism:** Background scheduler checks active challenge end timestamps.
- **Automated Transition:** When the countdown hits `00:00:00` , challenge automatically transitions to `COMPLETED` , freezes logging forms, and executes the punishment calculation routine.

§4.4 Auto-Balancer Snake Draft & 15-Hour Cap [v1]

- **15-Hour Sanity Cap:** System rejects any declared target > 15 hours/day (> 105 hours/week).
- **Draft Algorithm:**
 - Calculates composite rating: $R_i = 0.6 \times \text{DeclaredTarget} + 0.4 \times \text{HistoricalDailyAverage}$.
 - Sorts participants and runs a snake draft ($A, B, B, A, A, B \dots$) to produce evenly matched teams.
 - **Host Override:** Drag-and-drop roster editor lets mods swap participants before launching.

§4.5 Event Lock & Finalize Results [P0]

- **Route:** /admin/challenges/:id
- **Action:** Host clicks "Lock Final Results". Freezes all participant data rows and triggers the final punishment evaluation.

5. Module Specifications: Pre-Challenge Declarations

§5.1 Declared Weekly Target Hours [P0]

- **Requirement:** Every enrolled participant enters their target hours in HH:MM:SS (e.g., 35h 00m 00s) during the UPCOMING phase.
- **Constraint:** Minimum 1 hour, maximum 105 hours per week.

§5.2 Mandatory Weekly Goals Checklist [P0]

- **Requirement:** Each participant must submit between 1 and 10 concrete text tasks (e.g., *"Finish Organic Chemistry ch 4–6"*).
- **Hard Condition:** Every declared task must be checked off before the challenge concludes to avoid punishment.

§5.3 Pre-Kickoff Declaration Lock [P0]

- **Invariant:** When the event status becomes ACTIVE , all targets and task descriptions become read-only for participants.

§5.4 Host Goal Unlock & Mid-Event Edit [P0]

- **Purpose:** Accommodate real-life syllabus shifts or illness.
- **Action:** Admins can open any participant's goal sheet to add, edit, or unlock a task mid-challenge.

6. Module Specifications: Study Hour Ingestion & Logging

§6.1 Daily Clock-Time Self-Logging (HH:MM:SS) [P0]

- **Route:** /dashboard
- **Input Fields:** Date selector + three numeric inputs: Hours , Minutes , Seconds .
- **Display Representation:** Stored as total integer seconds, rendered formatted as HH:MM:SS (matching YPT display).

§6.2 24-Hour Single-Day Limit Validation [P0]

- **Rule:** Total study time logged for any single participant on any single calendar date cannot exceed 86,400 seconds (24 hours).

§6.3 Direct YPT API Bot Ingestion Endpoint [P1]

- **Route:** POST /api/v1/ingest/ypt
- **Authentication:** Pre-shared Bearer API Token between teammate's YPT bot and web server.
- **Payload:** { discordId, date, durationSeconds, yptSubject } .
- **Behavior:** Automatically upserts DailyStudyLog records and recalculates standings without human effort.

§6.4 Admin Inline Hours Override Grid [P0]

- **Route:** /admin/challenges/:id/roster
- **Capability:** Full table of all member logs with inline-editable hours to fix timer crashes or disputes. Logs flagged with is_admin_override = true .

7. Module Specifications: Standings, Scoreboard & Calculations

§7.1 Head-to-Head Live Match Scoreboard [P0]

- **Layout:** High-contrast top banner displaying Team A vs Team B (e.g., *Bees vs Butterflies*).

- **Metrics:** Total cumulative time (HH:MM:SS), leader crown icon, and lead margin delta (+Xh Ym Zs ahead).

§7.2 Unified Roster Standings Table [P0]

- **Columns:** Rank, Participant (Avatar + Discord Handle), Team Tag, Total Logged (HH:MM:SS), Target (HH:MM:SS), % Completed, Goals Done (M/N), Status Badge (On Track / At Risk).
- **Filters:** "All", "Team A", "Team B".

§7.3 Dynamic Daily Catch-Up Deficit Engine [P0]

- **Formula:**

$$\text{Deficit} = \max(0, \text{Target Seconds} - \text{Logged Seconds})$$

$$\text{Required Pace / Day} = \frac{\text{Deficit}}{\text{Days Remaining}}$$

- **UI Feedback:** Displays dynamic encouragement: *"Need 3h 45m/day over next 2 days to pass target."*

§7.4 Overflow Diminishing Returns Scoring Engine [v1]

- **Problem Solved:** Prevents single extreme outliers from breaking team competitive balance.
- **Dual-Credit Invariant:**
 - Personal Profile / Badges:** Always awards **100% full credit** for all logged hours.
 - Team Match Score:** Hours beyond daily target apply a half-weight damping factor ($\alpha = 0.5$):

$$\text{If } L \leq T_{\text{daily}} : \quad \text{TeamScore} = L$$

$$\text{If } L > T_{\text{daily}} : \quad \text{TeamScore} = T_{\text{daily}} + 0.5 \times (L - T_{\text{daily}})$$

8. Module Specifications: Accountability & Punishment

§8.1 Dual-Failure Auto-Flagging Engine [P0]

- **Trigger:** Evaluated automatically upon event conclusion:

$$\text{Is Punished} = (\text{Logged Seconds} < \text{Target Seconds}) \vee (\text{Incomplete Goals} > 0)$$

- **Result:** Failed participants assigned status `PUNISHED` .

§8.2 Punishment Wall & Deficit Roster [P0]

- **Display:** High-visibility section showing flagged members, their missing hours deficit, and unfinished tasks.

§8.3 Direct Punishment PFP Asset Download [P0]

- **Action:** Prominent "**Download Punishment PFP**" button on the Punishment Wall. Downloads the challenge PFP directly so members don't have to hunt Discord chat channels.

§8.4 Host Pardon / Excuse Override [P0]

- **Action:** Admin review modal allowing hosts to grant pardon (`EXCUSED`) with an audit reason (e.g., illness).

§8.5 Hall of Accountability Historical Archive [v1]

- **Description:** Permanent record tracking total punishments incurred, target completion rate, and redemption history across past challenges.

9. Module Specifications: Discord Broadcaster & Bot

§9.1 1-Click Formatted Markdown Summary Copy [P0]

- **Location:** Admin Console.
- **Action:** One-click button copies formatted Discord markdown to clipboard containing:

- Event title & dates.
- Final team scores & winning team announcement.
- Individual podium (🏆 , 🥈 , 🥉).
- Punishment Wall roster with Discord mentions.

§9.2 24/7 @HoIdMeToItBot Slash Commands [P1]

- **Commands:**
 - `/stats [user]` : Shows daily logged time, weekly target progress, and goal status.
 - `/standings` : Returns live team scoreboard and podium rankings.
 - `/deficit` : Displays remaining hours and required daily catch-up pace.

§9.3 Automated Daily & Final Results Embeds [P1]

- **Daily Check-in:** Bot posts a 12:00 PM mid-day standings embed to `#study-announcements` .
- **Final Broadcast:** Bot automatically posts the final winner/punishment embed when the countdown timer expires.

10. Module Specifications: Gamification, Streaks & Badges

§10.1 Daily Challenge Streaks [v1]

- **Rule:** Daily streak increments by 1 if a participant logs $\geq 100\%$ of their required daily target ($T_{\text{weekly}}/7$). Resets if a challenge day ends with 0 hours.

§10.2 Condition-Based Achievement Badges [v1]

Badge Name	Icon	Trigger Condition
Centurion		Log ≥ 100 hours of verified study time across challenges.
Night Owl		Log ≥ 30 hours between 10:00 PM and 4:00 AM.
Flawless Grinder		Complete 100% of target hours AND 100% of weekly goals in a challenge.
Comeback Kid		Overcome a > 5 -hour deficit in the final 48 hours to meet target.

Badge Name	Icon	Trigger Condition
Veteran		Participate in 5 completed community challenge events.

§10.3 Member Profile Cockpit (/profile/[id]) [v1]

- **Route:** /profile/[id]
- **Content:** Total lifetime hours, W/L record in team battles, current daily streak, badge showcase, and Hall of Accountability record.

11. Module Specifications: Peer-to-Peer Real-Time Features

§11.1 Spontaneous 1v1 Mutual Study Sprint Duels [v2]

- **Workflow:**
 - Member A initiates: /duel @user duration:2h .
 - Member B accepts via Discord button or web notification.
 - A temporary head-to-head sprint room is created with live countdown.
 - Winner declared automatically when the duration expires.
 - **Zero Admin Friction:** No moderator creation, review, or approval needed.

12. Deliberately Excluded / Rejected Features

The following features were evaluated and deliberately excluded based on user feedback:

- **✗ Native In-Browser Study Timer (Pomodoro/Stopwatch):** Excluded because members study on diverse mobile devices where YPT is preferred.
- **✗ In-App Virtual Study Rooms:** Excluded because community members already study inside Discord voice/video channels.
- **✗ Automated Discord Role Assignments (Roles for Winners/Losers):** Excluded for now to avoid server role clutter.
- **✗ Grace Passes / Freeze Days:** Excluded in favor of the pure cumulative catch-up deficit model.